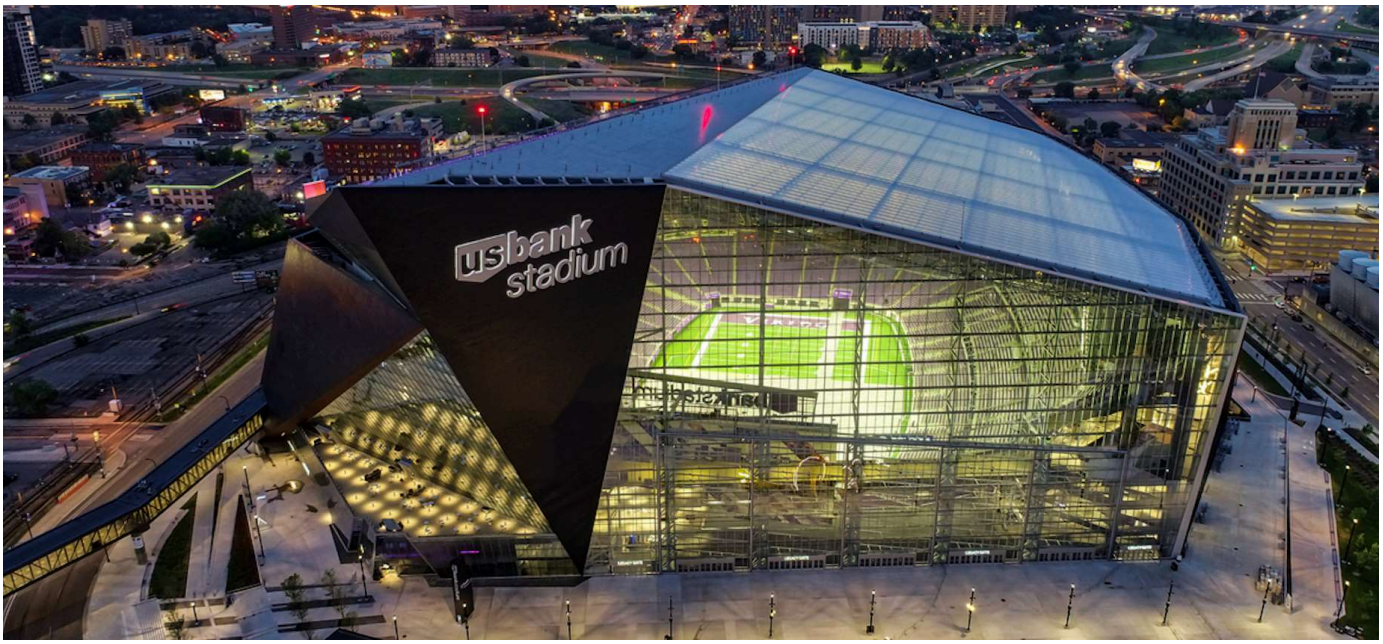


Whether or not you like football, the Super Bowl is a spectacle. There's a little something for everyone at your Super Bowl party. Drama in the form of blowouts, comebacks, and controversy for the sports fan. There are the ridiculously expensive ads, some hilarious, others gut-wrenching, thought-provoking, and weird. The half-time shows with the biggest musicians in the world, sometimes riding giant mechanical tigers or leaping from the roof of the stadium.

The dataset we'll use was scraped and polished from Wikipedia. It is made up of three CSV files, one with game data, one with TV data, and one with halftime musician data for 52 Super Bowls through 2018.

The Data

Three datasets have been provided, and summaries and previews of each are presented below.

1. halftime_musicians.csv

This dataset contains information about the musicians who performed during the halftime shows of various Super Bowl games. The structure is shown below, and it applies to all remaining files.

Column	Description
`'super_bowl'`	The Super Bowl number (e.g., 52 for Super Bowl LII).
`'musician'`	The name of the musician or musical group that performed during the halftime show.
`'num_songs'`	The number of songs performed by the musician or group during the halftime show.

2. super_bowls.csv

This dataset provides details about each Super Bowl game, including the date, location, participating teams, and scores, including the points difference between the winning and losing team (`'difference_pts'`).

3. tv.csv

This dataset contains television viewership statistics and advertisement costs related to each Super Bowl.

```
# Import libraries
import pandas as pd
from matplotlib import pyplot as plt
```

```python
# Load the CSV data into DataFrames
super_bowls = pd.read_csv("datasets/super_bowls.csv")
super_bowls.head()
```

	d...	s...	venue	city	st...	a...	team_winner
0	2018-02-04	52	U.S. Bank Stadium	Minneapolis	Minnesota	67612	Philadelphia Eagles
1	2017-02-05	51	NRG Stadium	Houston	Texas	70807	New England Patriot
2	2016-02-07	50	Levi's Stadium	Santa Clara	California	71088	Denver Broncos
3	2015-02-01	49	University of Phoenix Stadium	Glendale	Arizona	70288	New England Patriot
4	2014-02-02	48	MetLife Stadium	East Rutherford	New Jersey	82529	Seattle Seahawks

Rows: 5 Expand

```python
tv = pd.read_csv("datasets/tv.csv")
tv.head()
```

i...	super_b...	n...	avg_us_viewers	total_us_viewers	rating_household	share_househo
0	52	NBC	103390000		43.1	
1	51	Fox	111319000	172000000	45.3	
2	50	CBS	111864000	167000000	46.6	
3	49	NBC	114442000	168000000	47.5	
4	48	Fox	112191000	167000000	46.7	

Rows: 5 Expand

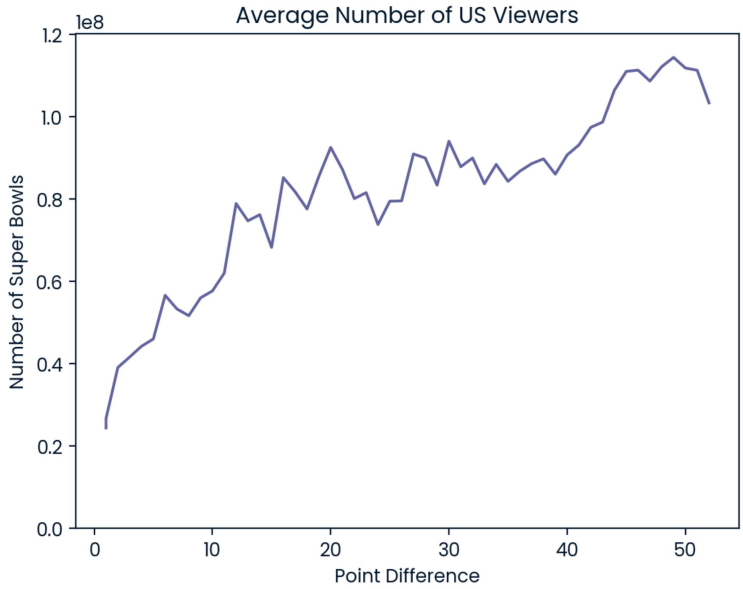
```python
halftime_musicians = pd.read_csv("datasets/halftime_musicians.csv")
halftime_musicians.head()
```

index	super_bowl	musician
0	52	Justin Timberlake
1	52	University of Minnesota Marching Band
2	51	Lady Gaga
3	50	Coldplay
4	50	Beyoncé

Rows: 5 Expand

```python
# Start coding here
# Use as many cells as you need
# Question 1
# Import pandas
import pandas as pd
from matplotlib import pyplot as plt

# Load the CSV data into DataFrames
tv = pd.read_csv("datasets/tv.csv")

# Display the TV data to inspect
tv.head()

# Find the year with the highest TV viewership
plt.plot(tv.super_bowl, tv.avg_us_viewers)
plt.title('Average Number of US Viewers')
viewership_increased = True
print(f"Super Bowl viewership increased over time.")

# Question 2

# Load the CSV data into DataFrames
super_bowls = pd.read_csv("datasets/super_bowls.csv")

# Display the Super Bowls data to inspect
super_bowls.head()

# Filter the data for point difference >40
difference = len(super_bowls[super_bowls["difference_pts"]>40])

# You can also plot a histogram of point differences to visualize the result
plt.hist(super_bowls.difference_pts)
plt.xlabel('Point Difference')
plt.ylabel('Number of Super Bowls')
plt.show()

print(f"The matches finishing with a point difference over 40 were: {difference}")

# Question 3
# Load the CSV data into DataFrames
halftime_musicians = pd.read_csv("datasets/halftime_musicians.csv")

# Display the halftime musicians data to inspect
halftime_musicians.head()

# Count halftime show songs for each musician
halftime_appearances = halftime_musicians.groupby('musician').sum('num_songs')
halftime_appearances = halftime_appearances.sort_values('num_songs', ascending=False)

most_songs = "Justin Timberlake"
```

```
Super Bowl viewership increased over time.
```




```
The matches finishing with a point difference over 40 were: 1
```